TASK 8: CAPTURE NETWORK TRAFFIC WITH WIRESHARK

Network traffic capture, display filters, packet inspection, and security observations

1. Objective

To capture live network traffic on a local Wi-Fi interface using Wireshark, apply display filters to isolate common protocols, inspect packet details, and record basic security observations.

2. Tools and Environment

Tool: Wireshark
Device/interface shown: Windows laptop, Wi-Fi capture interface
Capture file shown in Wireshark: Wi-Fi capture (.pcapng)
Capture summary visible in screenshots: approximately 1,399 packets; 0 dropped in the displayed capture summary.

3. Procedure Performed

1. Opened Wireshark and selected the Wi-Fi network interface.
2. Captured live packets while normal network activity was occurring.
3. Entered display filters in the filter bar to focus on specific traffic.
4. Selected packets and reviewed the packet-details and packet-bytes panes.
5. Captured screenshots as evidence of the filters and observed protocols.

4. Display Filters Used

Display filter	Purpose
`http`	Displays HTTP requests and responses.
`dns`	Displays DNS queries and responses.
`tcp`	Displays TCP packets.
`tcp.port == 80`	Displays TCP traffic where either port is 80.
`tcp.flags.syn == 1`	Displays TCP packets with the SYN flag set, commonly used to begin connections.
`tcp.stream == 0`	Displays packets in the selected TCP stream (stream number shown in the screenshot).

5. Observations and Security Notes

• The capture contains DNS queries and responses, indicating domain-name lookups.
• HTTP packets include GET requests for `/ncc.txt` and responses marked `HTTP/1.1 200 OK`, showing a successful HTTP response in the captured traffic.
• TCP packets show connection setup and acknowledgement activity. Some traffic is identified as TLS Application Data, which is encrypted at the application layer and is not readable as ordinary HTTP content in this capture view.
• HTTP is not encrypted by itself; request paths and headers may be visible to someone able to capture the traffic.

HTTPS should be used for sensitive communications.
• The screenshots include routine network traffic from the device. No claim is made that every packet or destination is malicious; packet capture alone does not prove an attack.

6. Conclusion

Wireshark was used to capture live Wi-Fi traffic and inspect packets with HTTP, DNS, TCP, port 80, SYN-flag, and TCP-stream display filters. The screenshots provide evidence of packet capture and protocol filtering. This exercise demonstrates how protocol filters and packet details can support basic network troubleshooting and security analysis.

7. Screenshot Evidence

The following pages contain screenshots supplied for this report. They show the filter bar, packet list, and packet details used during analysis.

Screenshot Evidence 1

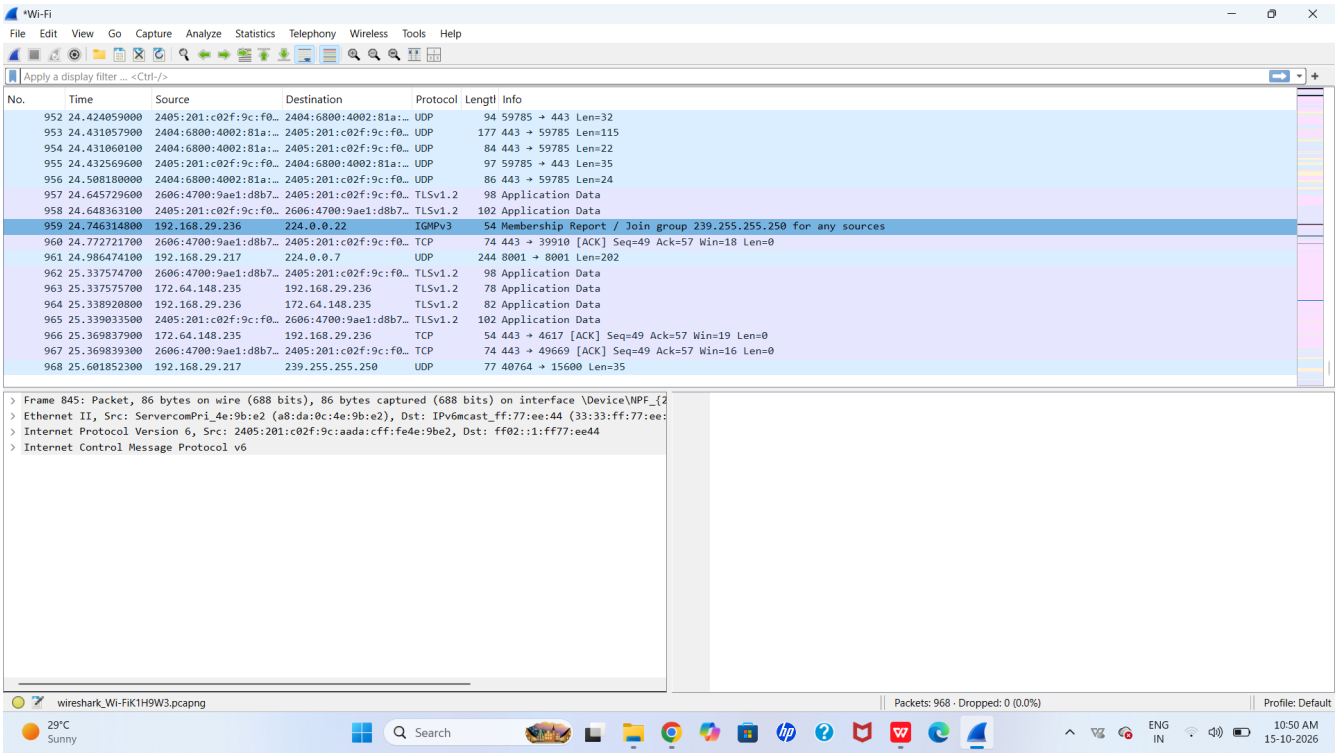


Figure 1. Captured packets and IPv6 network traffic.

Screenshot Evidence 2

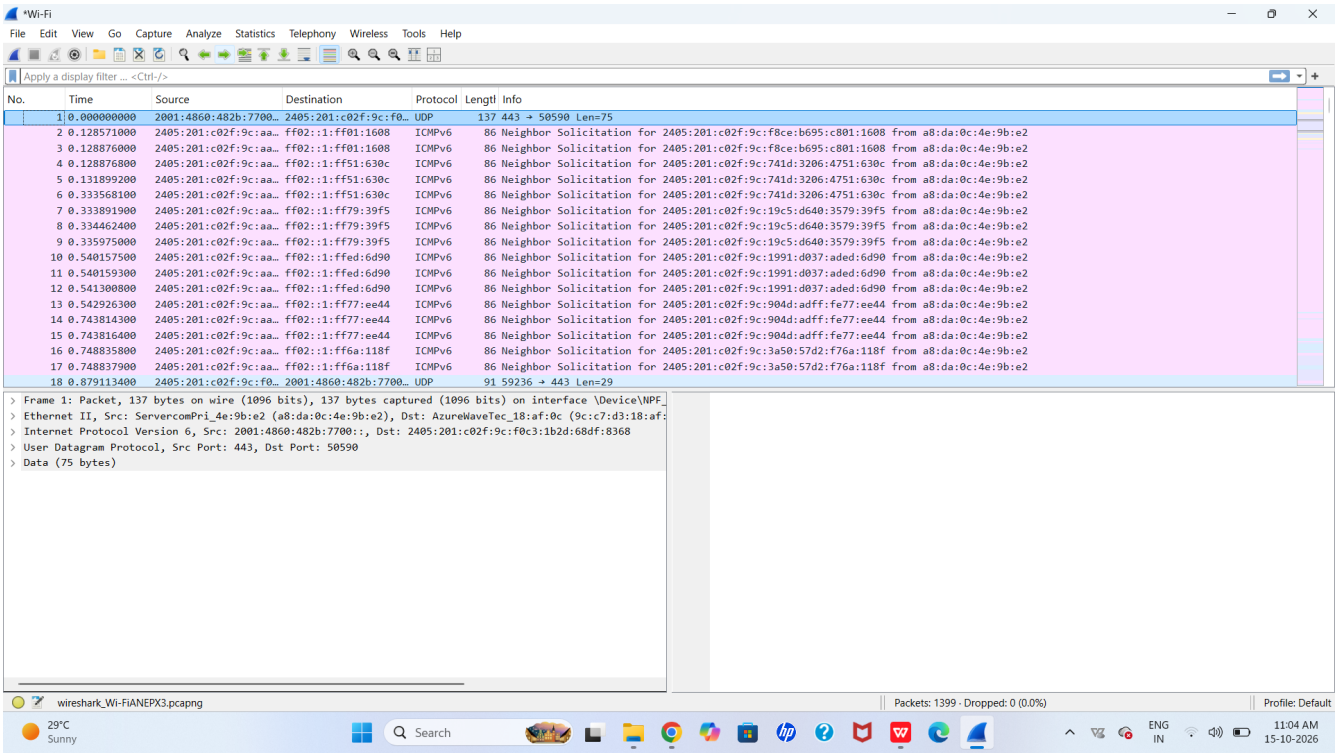


Figure 2. DNS filter showing DNS queries and responses, including domain lookups.

Screenshot Evidence 3

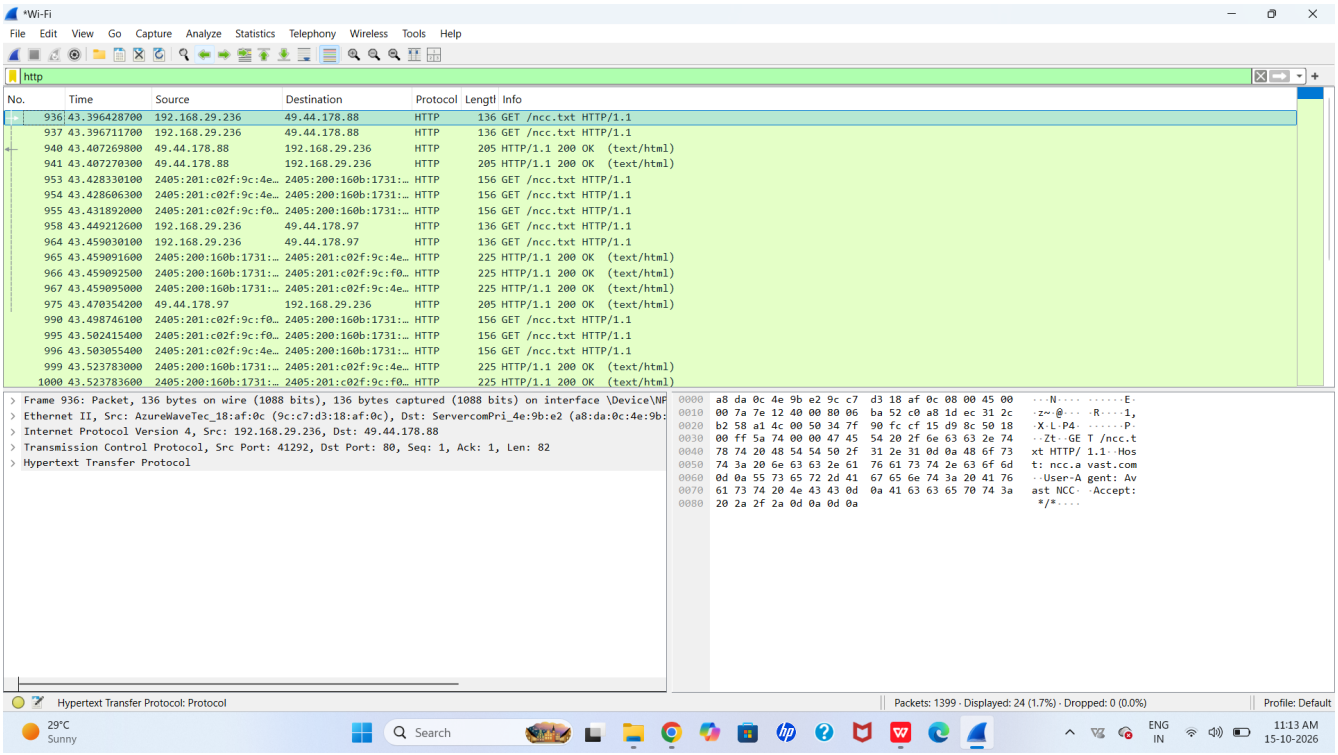


Figure 3. HTTP display filter showing GET requests and HTTP 200 OK responses.

Screenshot Evidence 4

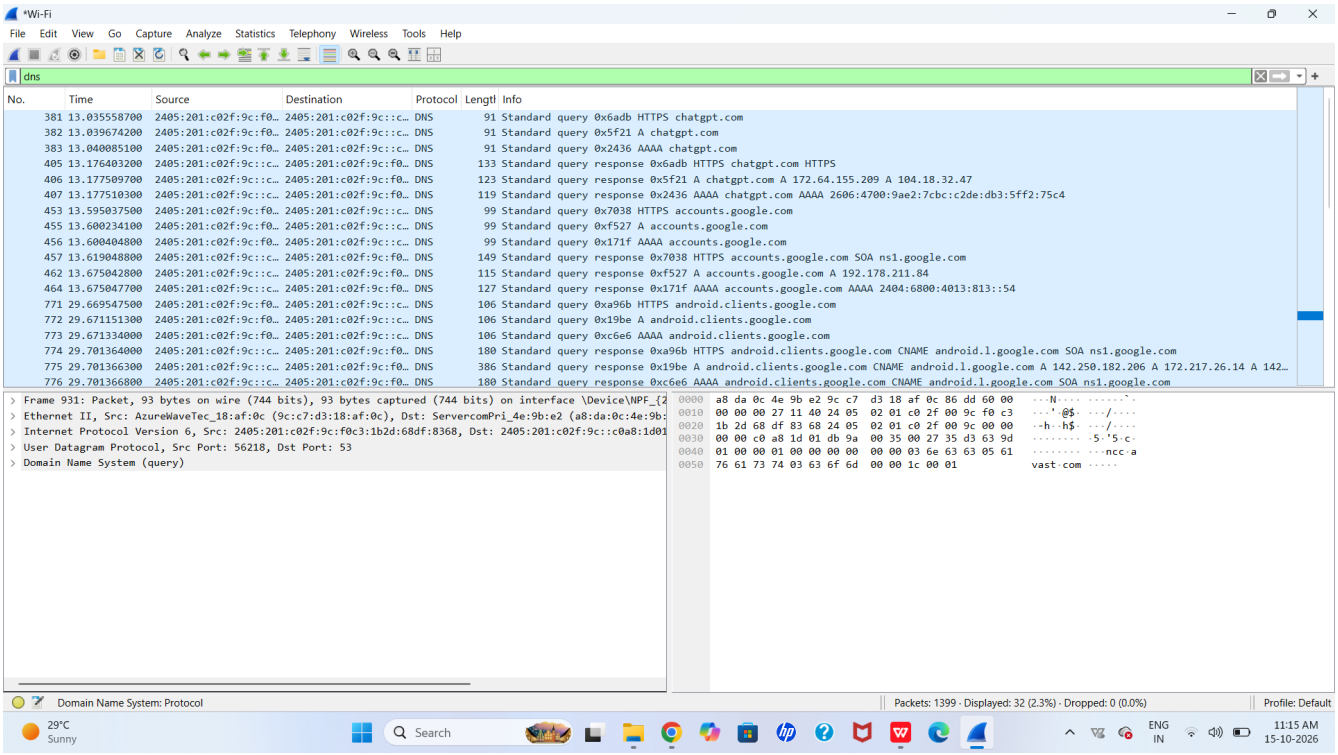


Figure 4. TCP filter showing TCP packets and TLS application-data traffic.

Screenshot Evidence 5

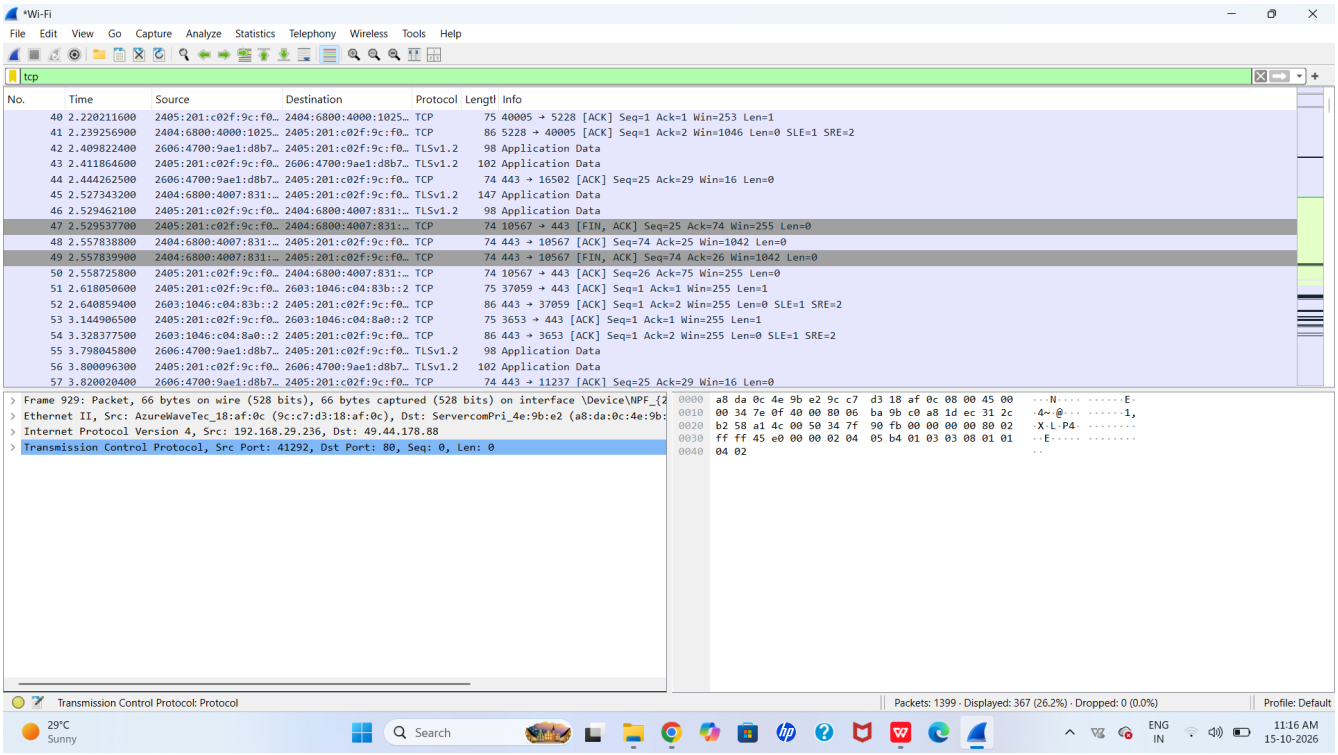


Figure 5. TCP SYN filter showing connection-establishment packets.

Screenshot Evidence 6

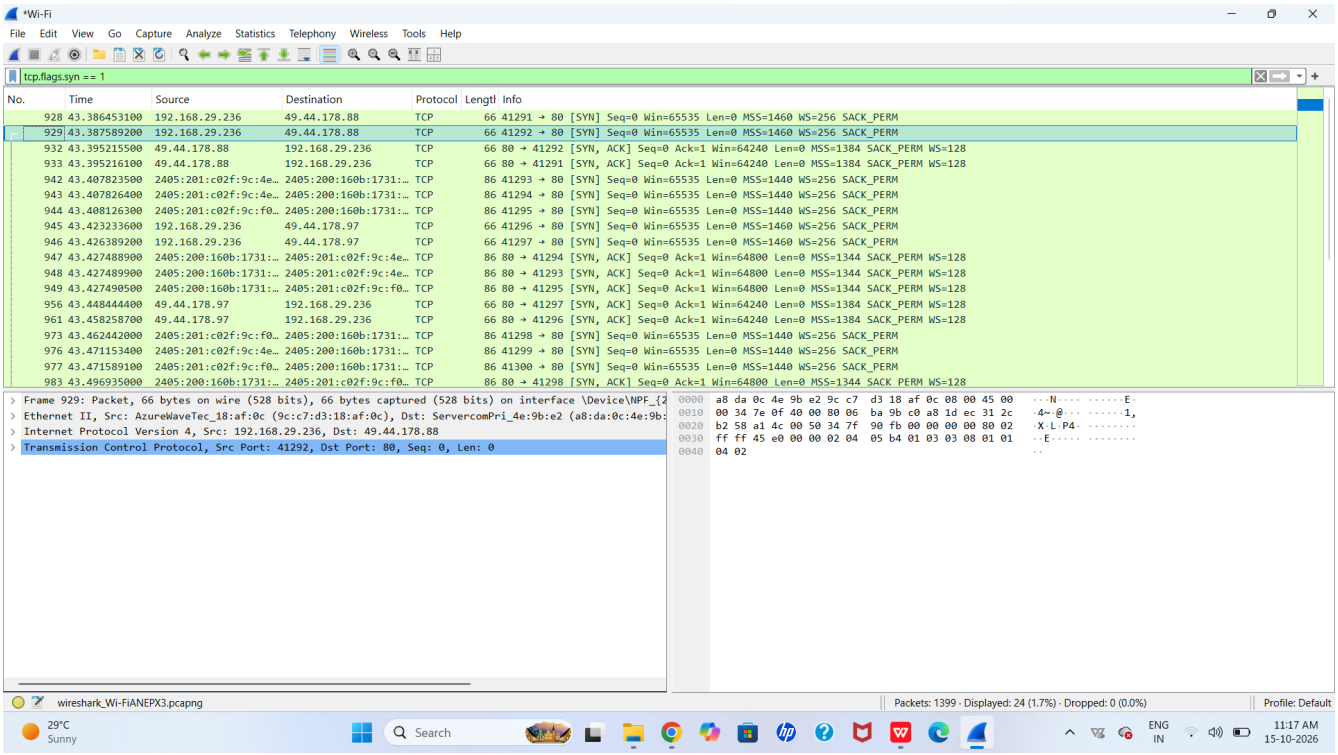


Figure 6. TCP port 80 filter showing HTTP traffic on port 80.

Screenshot Evidence 7

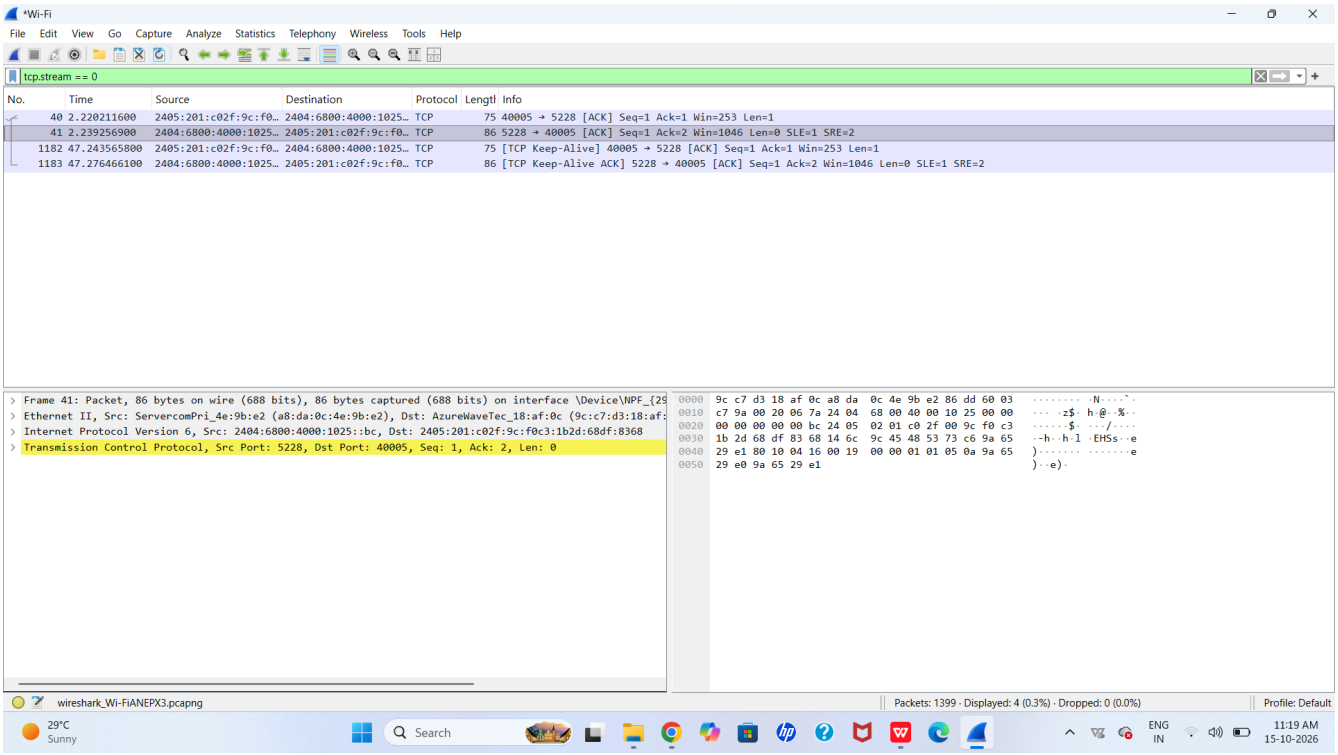


Figure 7. TCP stream filter showing packets belonging to one selected stream.

Screenshot Evidence 8

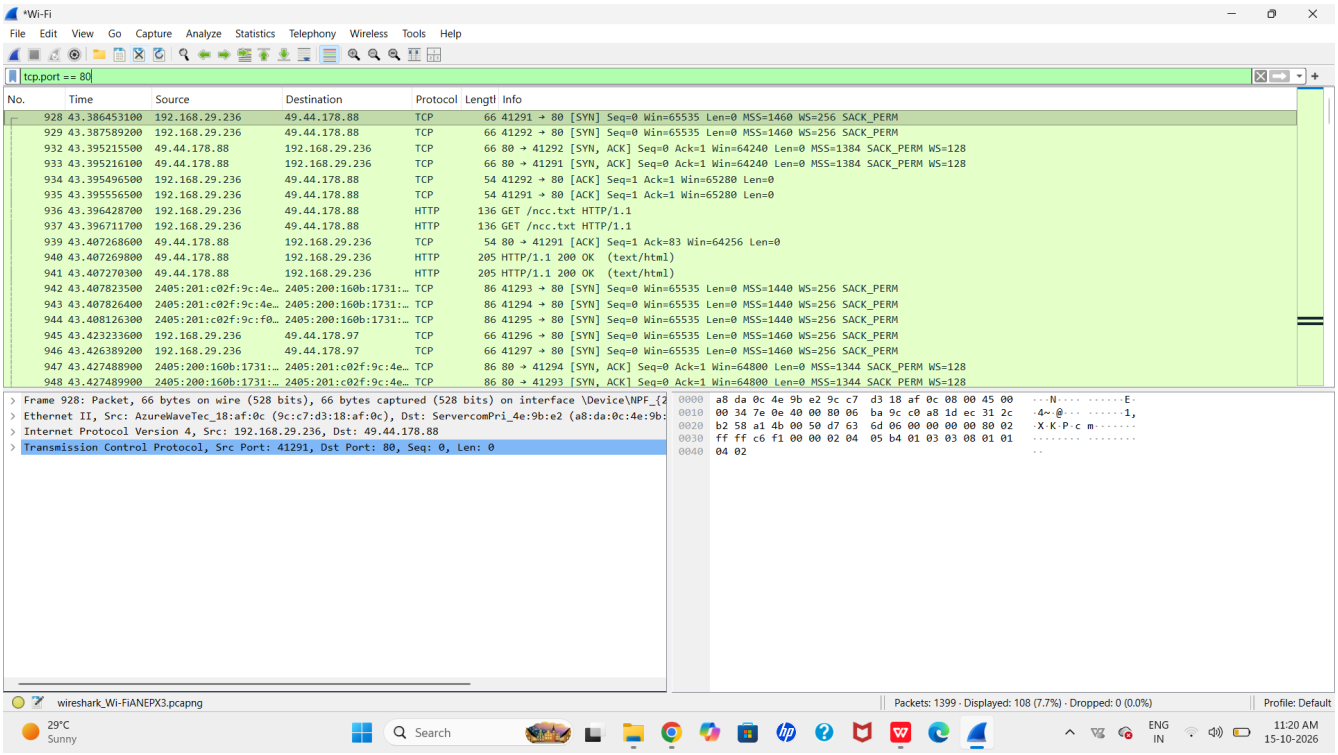


Figure 8. Additional captured traffic with protocol details visible.